

The following is a school assignment, in which a classmate and I conduct dynamic testing on a web-based drawing tool

Testing Approach

Dynamic black box testing was performed by toggling combinations of state tests, such as using the brush tool in light mode using a pattern and a set diameter and opacity values. Within each state test, repeated tool state combinations were conducted while changing one of the state values. For instance, to test the brush, the same light/paint setting, brush color, and diameter states would be toggled; however, the opacity states would be tested in accordance with the following boundary conditions: at the lowermost limit, at a commonly-used or popular opacity, and at the uppermost limit.

When some error states were found, the tester reloaded the page since there appeared to be no way to undo a mistake, such as selecting a cropped area by mistake. Others, such as selecting an “invalid” or blank color, were reversible and could be reversed by selecting a color from the History tab.

Equivalence Partitions

Color Equivalence Partitions:

Instead of testing every color individually, the sliders will be tested in groups. In addition, the slider values tested will also be grouped into minimums, maximums and middles. The sliders pertaining to hue (H, R, B, and G) will be tested together. Sliders V and S, which control brightness, are to also be tested together.

Gradient Equivalence Partitions:

Each slider will be tested in groups of their maximums, minimums, and middle values.

Pattern Equivalence Partitions:

The opacity slider cases will be in groups of maximums, minimums, and middle values.

Options Partitions:

Every slider under the options tab will be tested in groups of maximums, minimums, and middle values.

Devices and Browsers Partitions:

Instead of testing every device and browser, we will test devices in groups of apple, android, and windows. Browsers will be grouped into google browsers, apple browsers, microsoft browsers and a couple randoms.

Boundary Conditions

Selectors that do not contain sliders or ways of input will be unavailable to apply boundary conditions.

Color Boundaries:

The color tab has multiple sliders. For our expected-to-pass test cases we will test the maximums, minimums, and middles of each slider. The hue slider will be tested at 0 and at 360. The R, G, and B sliders will be tested at 0 and 255. The value, saturation, and opacity sliders will be tested at 0 and 100. They will also each be tested at their respective middle values. For our expected-to-fail test cases we will test input one below each minimum and one above each maximum.

Gradient Boundaries:

The gradient tab has 3 sliders which will be tested expected-to-pass at their maximum, minimum, and middle boundaries. The opacity slider will be tested at 0, 100 and 50. The colors slider will be tested as far left as they will go, far right as they will go and middle. The circle slider is difficult to test boundaries as there is no clear high or low. Instead, it will be tested at 95 and 270 degrees. For our expected-to-fail test cases we will test input for opacity at -1 and 101.

Pattern Boundaries:

The pattern tab has one opacity slider. The expected-to-pass tested boundaries will be at 0, 100, and 50. For our expected-to-fail test cases we will test input at -1 and 101.

Options Boundaries:

The options have multiple subsections for pencil, brush, and calligraphy. The pencil and calligraphy sections will be expected-to-pass tested with their slider's boundaries and middles for diameter and opacity. Diameter for pencil has a minimum of 1, a maximum of 15 and a middle of 7. For calligraphy, the diameter boundaries are at 1 and 100 with a middle at 50. Both have opacity boundaries at 1 and 100 and middles at 50. The brush tab has four sliders for diameter, hardness, flow, and opacity. Each slider will be tested at their low of 1, high of 100, and middle of 50. For our expected-to-fail test cases we will test input one below each minimum and one above each maximum.

Test Cases

Paint Tab

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Paint Tab	All tools on left proceed with selected paint color	All tools on left proceed with selected paint color	Pass	

Light Tab

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Light Tab	All tools on left proceed in white with selected paint color border	All tools on left proceed in white with selected paint color border	Pass	

Color Tab

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Color Tab	Upon selection, color window opens	Upon selection, color window opens	Pass	
Color Tab- Hues	Color selection slider (360-0), R, G and B (255-0) sliders coinciding with selected hue	Color selection slider (360-0), R, G and B (255-0) sliders coinciding with selected hue	Pass	
Color Tab- S, V	Sliders that adjusts the brightness and darkness of the selected hue (100-0)	Sliders that adjusts the brightness and darkness of the selected hue (100-0)	Pass	
Color Tab- A	Slider that adjusts the opacity of the selected color (0-100)	Slider that adjusts the opacity of the selected color (0-100)	Pass	
Color Tab- All Sliders Set At 0	No color	No color	Pass	

Color Tab- All Sliders Set At Max	Red	Sliders auto adjust, color combination impossible	Fail	The sliders will automatically adjust themselves and it is impossible to max all of them at once
Color Tab- All Sliders Set At Middle	Semi-transparent gray	Sliders auto adjust, color combination impossible	Fail	The sliders will automatically adjust themselves and it is impossible for all of them to be at the middle at once
Color Tab- All Sliders Set Below 0	Not possible	Not possible	Pass	
Color Tab- All Sliders Set Above Maximum	Not possible	Not possible	Pass	

Gradient Tab

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Gradient Tab	Upon selection, gradient window opens	Upon selection, gradient window opens	Pass	
Gradient Tab- Middle Slider	Slider adjusts opacity of color	Slider adjusts opacity of color	Pass	
Gradient Tab- Middle Slider 100	Solid color, no transparency	Solid color, no transparency	Pass	
Gradient Tab- Middle Slider 0	Fully transparent, no color	Fully transparent, no color	Pass	
Gradient Tab- Middle Slider 50	Semi transparent	Semi transparent	Pass	

Gradient Tab-Middle Slider 101	Not possible	Not possible	Pass	
Gradient Tab-Middle Slider -1	Not possible	Not possible	Pass	
Gradient Tab-Circle Slider	Slider adjusts the direction of gradient	Slider adjusts the direction of gradient	Pass	
Gradient Tab-Circle Slider Top of Circle	Slider at 90 degrees, left most color at top	Slider at 92 degrees, left most color at top	Fail	Degrees are off center, colors are in expected position
Gradient Tab-Circle Slider Bottom of Circle	Slider at 270 degrees, right most color at top	Slider at 267 degrees, right most color at top	Fail	Degrees are off center, colors are in expected position
Gradient Tab-Two Bottom Boxes	Selections for the colors of the gradient	Selections for the colors of the gradient	Pass	
Gradient Tab-Two Bottom Sliders	Sliders control how much of each selected color are present in the gradient, how harsh the blend between is	Sliders control how much of each selected color are present in the gradient, how harsh the blend between is	Pass	
Gradient Tab-Two Bottom Sliders Full Right	Only left color present in gradient	Only left color present in gradient	Pass	
Gradient Tab-Two Bottom Sliders Full Left	Only right color present in gradient	Only right color present in gradient	Pass	
Gradient Tab-Two Bottom Sliders Both Middle	Half and half colors, no blend	Half and half colors, no blend	Pass	

Gradient Tab- Additional Gradient Colors	Click between or beside other boxes to add more gradient colors	Click between or beside other boxes to add more gradient colors	Pass	
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Pattern Tab

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Pattern Tab	Upon selection, pattern window opens	Upon selection, pattern window opens	Pass	
Pattern Tab- Slider 100	Completely opaque pattern	Completely opaque pattern	Pass	
Pattern Tab- Slider 0	Completely transparent pattern	Completely transparent pattern	Pass	
Pattern Tab- Slider 50	Semi transparent pattern	Semi transparent pattern	Pass	
Pattern Tab- Slider -1	Not possible	Not possible	Pass	
Pattern Tab- Slider 101	Not possible	Not possible	Pass	

Swatch Tab

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Swatch Tab	Upon selection, swatch window opens	Upon selection, swatch window opens	Pass	
Swatch Tab- Solids	Swatch of solid colors is displayed	Swatch of solid colors is displayed	Pass	
Swatch Tab- Solids Selector	A selector of color schemes is present	A selector of color schemes is present	Pass	

Swatch Tab-Gradients	Swatch of gradient colors is displayed	Swatch of gradient and solid colors is displayed	Fail	Not all colors are gradient, some are solid
Swatch Tab-Gradient Selector	A selector of gradient color schemes is present	A selector of gradient color schemes is present	Pass	
Swatch Tab-Patterns	Swatch of patterns is displayed	Swatch of patterns is displayed	Pass	
Swatch Tab-Pattern Selector	A selector of pattern schemes is present	A selector of pattern schemes is present	Pass	

History Tab

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
History Tab	Upon selection, history window opens, showing last 15 operations	Upon selection, history window opens, showing last 15 operations	Pass	
History Tab-Titles	Each operation is labeled as what it was	Operations are incorrectly labeled	Fail	Some operations were given random titles, eraser for paint, pattern for eraser, etc.

Options Tab

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Options Tab	Upon selection, options window opens	Upon selection, options window opens	Pass	

Options Tab - Pencil

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
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Options Tab- Pencil Selector	Pencil selector displayed with diameter and opacity slider	Pencil selector displayed with diameter and opacity slider	Pass	
Options Tab- Pencil Selector Diameter Slider Max (15)	Pencil draws with wide diameter	Pencil draws with wide diameter	Pass	
Options Tab- Pencil Selector Diameter Slider Min (1)	Pencil draws with thin diameter	Pencil draws with thin diameter	Pass	
Options Tab- Pencil Selector Diameter Slider Middle (7)	Middle diameter between max and min	Middle diameter between max and min	Pass	
Options Tab- Pencil Selector Diameter Slider 16	Not possible	Not possible	Pass	
Options Tab- Pencil Selector Diameter Slider 0	Not possible	Not possible	Pass	
Options Tab- Pencil Selector Opacity Slider Max (100)	Solid, non transparent pencil	Solid, non transparent pencil	Pass	
Options Tab- Pencil Selector Opacity Slider Min (1)	Mostly transparent pencil	Mostly transparent pencil	Pass	
Options Tab- Pencil Selector Opacity Slider Middle (50)	Semi transparent pencil	Semi transparent pencil	Pass	

Options Tab- Pencil Selector Opacity Slider 101	Not possible	Not possible	Pass	
Options Tab- Pencil Selector Opacity Slider 0	Not possible	Not possible	Pass	
Options Tab - Brush Selector Blank Color	Not Possible	Not Possible	Pass	Only possible after selecting an erased section

Options Tab - Brush

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Options Tab- Brush Selector	Brush selector displayed with diameter and opacity slider	Brush selector displayed with diameter and opacity slider	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Diameter Slider Max (100)	Brush draws with wide diameter	Brush draws with wide diameter	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Diameter Slider Min (1)	Brush draws with thin diameter	Brush draws with thin diameter	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Diameter Slider Middle (50)	Brush draws with diameter between max and min	Brush draws with diameter between max and min	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Diameter Slider 101	Not possible	Not possible	Pass	Later, upon reloading the page the brush option opened a blank window

Options Tab- Brush Selector Diameter Slider 0	Not possible	Not possible	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Opacity Slider Max (100)	Solid, non transparent brush	Solid, non transparent brush	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Opacity Slider Min (1)	Mostly transparent brush	Mostly transparent brush	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Opacity Slider Middle (50)	Semi transparent brush	Semi transparent brush	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Opacity Slider 101	Not possible	Not possible	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Opacity Slider -1	Not possible	Not possible	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Hardness Slider Max (100)	Brush draws with hard strokes	Brush draws with hard strokes	Pass	Later, upon reloading the page the brush option opened a blank window

Options Tab- Brush Selector Hardness Slider Min (1)	Brush draws with soft strokes	Brush draws with soft strokes	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Hardness Slider Middle (50)	Brush draws with medium soft/hard strokes	Brush draws with medium soft/hard strokes	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Hardness Slider 101	Not possible	Not possible	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Hardness Slider 0	Not possible	Not possible	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Flow Slider Max (100)	Smooth brush flow	Smooth brush flow	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Flow Slider Min (1)	Brush strokes are disconnected or bumpy	Brush strokes are disconnected or bumpy	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Flow Slider Middle (50)	Brush strokes are medium smooth/ bumpy	Brush strokes are medium smooth/ bumpy	Pass	Later, upon reloading the page the brush option opened a blank window

Options Tab- Brush Selector Flow Slider 101	Not possible	Not possible	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab- Brush Selector Flow Slider 0	Not possible	Not possible	Pass	Later, upon reloading the page the brush option opened a blank window
Options Tab - Brush Selector Blank Color	Not Possible	Not Possible	Pass	Only possible after selecting an erased section

Options Tab - Calligraphy

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Options Tab- Calligraphy Selector	Calligraphy selector displayed with diameter and opacity slider	Calligraphy selector displayed with diameter and opacity slider	Pass	
Options Tab- Calligraphy Selector Diameter Slider Max (100)	Calligraphy draws with wide diameter	Calligraphy draws with wide diameter	Pass	
Options Tab- Calligraphy Selector Diameter Slider Min (1)	Calligraphy draws with thin diameter	Calligraphy is not visible	Fail	Calligraphy is not visible at all
Options Tab- Calligraphy Selector Diameter Slider Middle (50)	Calligraphy draws with diameter between max and min	Calligraphy draws with diameter between max and min	Pass	

Options Tab- Calligraphy Selector Diameter Slider 101	Not possible	Not possible	Pass	
Options Tab- Calligraphy Selector Diameter Slider 0	Not possible	Not possible	Pass	
Options Tab- Calligraphy Selector Opacity Slider Max (100)	Solid, non transparent calligraphy	Solid, non transparent calligraphy	Pass	
Options Tab- Calligraphy Selector Opacity Slider Min (1)	Mostly transparent calligraphy	Mostly transparent calligraphy	Pass	
Options Tab- Calligraphy Selector Opacity Slider Middle (50)	Semi transparent calligraphy	Semi transparent calligraphy	Pass	
Options Tab- Calligraphy Selector Opacity Slider 101	Not possible	Not possible	Pass	
Options Tab- Calligraphy Selector Opacity Slider 0	Not possible	Not possible	Pass	
Options Tab - Calligraphy Selector Blank Color	Not Possible	Not Possible	Pass	Only possible after selecting an erased section

Save Button

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
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Save Button	Popup asking for filename and type	No response	Fail	Button non responsive/ works as intended on FireFox browser
Save as JPEG	Artwork saves as JPEG	Not applicable	Fail	JPEG option not available/ works as intended on FireFox browser
Save as PNG	Artwork saves as PNG	Not applicable	Fail	PNG option not available/ works as intended on FireFox browser
Save as SVG	Artwork saves as SVG	Not applicable	Fail	SVG option not available/ works as intended on FireFox browser
Save as PDF	Artwork saves as PDF	Not applicable	Fail	PDF option not available/ works as intended on FireFox browser

Tools Tab - Marquee

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Options Tab - Ellipses	Select an ellipsis-shaped area within the canvas within the bounds determined by the user	Select an ovular-shaped area within the canvas within the bounds determined by the user	Fail	Tool works properly, but the name may be deceptive Compatible with all tools except for "Fill"
Options Tab - Polygon	Select an polygon-shaped area within the canvas within the bounds determined by the user	Select an polygon-shaped area within the canvas within the bounds determined by the user	Fail	Compatible with all tools except for "Fill"

Options Tab - Star	Select an “star”-shaped area within the canvas within the bounds determined by the user	Select an “star”-shaped area within the canvas within the bounds determined by the user	Fail	Compatible with all tools except for “Fill”
Options Tab - Burst	Select an “burst”-shaped area within the canvas within the bounds determined by the user	Select an “burst”-shaped area within the canvas within the bounds determined by the user	Fail	Compatible with all tools except for “Fill”
Options Tab - Gear	Select a “gear”-shaped area within the canvas within the bounds determined by the user	Select a “gear”-shaped area within the canvas within the bounds determined by the user	Fail	Compatible with all tools except for “Fill”

Tools Tab - Crop

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Display (1440x900)	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
Original (700x575)	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
2x3	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
3x5	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping

4x3 (DVD)	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
4x3 (Book)	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
4x6 (Postcard)	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
5x7 (L, 2L)	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
8x10	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
16x9 (HD)	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
16x20	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
20x30 (Poster)	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping
Square	Select and crop the selected area	Select and crop the selected area	Pass	There is no way to “undo” after cropping

Tools Tab - Text

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Text Size 0, text field filled	No text visible	Small opaque speech bubble visible	Pass	Technical pass; couldn't see text
Text Size 130, text field filled	Medium text visible	Medium text visible	Pass	Technical pass; couldn't see text

Text Size 256, text field filled	Very large text visible	Very large text visible	Pass	
Stroke 0, text field filled	Very light stroke around text	Very light stroke around text	Pass	Stroke cannot be transparent
Stroke 10, text field filled	Medium stroke around text	Medium stroke around text	Pass	Stroke cannot be transparent
Stroke 20, text field filled	Very heavy stroke around text	Very heavy stroke around text	Pass	Stroke cannot be transparent

Tools Tab - Shape

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Ellipses, Anchored, Stroke 1	Create ellipses shape with a light stroke weight	Create ovular shape with a light stroke weight	Pass	Tool works properly, but the name may be deceptive
Ellipses, Anchored, Stroke 50	Create ellipses shape with a medium stroke weight	Create ovular shape with a medium stroke weight	Pass	Tool works properly, but the name may be deceptive
Ellipses, Anchored, Stroke 100	Create ellipses shape with a heavy stroke weight	Create ovular shape with a heavy stroke weight	Pass	Tool works properly, but the name may be deceptive
Ellipses, Freedraw, Stroke 1	Create ellipses path with a light stroke weight	Create ovular path with a light stroke weight	Pass	Tool works properly, but the name may be deceptive
Ellipses, Freedraw, Stroke 50	Create ellipses path with a medium stroke weight	Create ovular path with a medium stroke weight	Pass	Tool works properly, but the name may be deceptive
Ellipses, Freedraw, Stroke 100	Create ellipses path with a heavy stroke weight	Create ovular path with a heavy stroke weight	Pass	Tool works properly, but the name may be deceptive

Ellipses, Active, Stroke 1	Repeated ellipses pattern in selected area with light stroke weight	Repeated ovular pattern in selected area with light stroke weight	Pass	Tool works properly, but the name may be deceptive
Ellipses, Active, Stroke 50	Repeated ellipses pattern in selected area with medium stroke weight	Repeated ovular pattern in selected area with medium stroke weight	Pass	Tool works properly, but the name may be deceptive
Ellipses, Active, Stroke 100	Repeated ellipses pattern in selected area with heavy stroke weight	Repeated ovular pattern in selected area with heavy stroke weight	Pass	Tool works properly, but the name may be deceptive
Polygon, Anchored, Sides 2, Stroke 1	Create polygon shape with the expected sides, a light stroke weight, and an expected corner condition	Create polygon shape with the expected sides and a light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Anchored, Sides 50, Stroke 1	Create polygon shape with the expected sides, a light stroke weight, and an expected corner condition	Create polygon shape with the expected sides and a light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Anchored, Sides 100, Stroke 1	Create polygon shape with the expected sides, a light stroke weight, and an and the expected corner condition	Create polygon shape with the expected sides and a light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance

Polygon, Anchored, Stroke 50	Create polygon shape with a medium stroke weight, and an expected corner condition	Create polygon shape with a medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Anchored, Sides 50, Stroke 50	Create polygon shape with the expected sides, a medium stroke weight, and an expected corner condition	Create polygon shape with the expected sides and a medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Anchored, Sides 100, Stroke 50	Create polygon shape with the expected sides, medium stroke weight, and an expected corner condition	Create polygon shape with the expected sides and a medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Anchored, Sides 2, Stroke 100	Create polygon shape with the expected sides, heavy stroke weight, and an expected corner condition	Create polygon shape with the expected sides and a heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Anchored, Sides 50, Stroke 100	Create polygon shape with the expected sides, heavy stroke weight, and an expected corner condition	Create polygon shape with the expected sides and a heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance

Polygon, Anchored, Sides 100, Stroke 100	Create polygon shape with the expected sides, heavy stroke weight, and an expected corner condition	Create polygon shape with the expected sides and a heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Freedraw, Sides 2, Stroke 1	Create polygon path with the expected sides, a light stroke weight, and an expected corner condition	Create polygon path with a light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Freedraw, Sides 50, Stroke 1	Create polygon path with the expected sides, a light stroke weight, and an expected corner condition	Create polygon path with a light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Freedraw, Sides 100, Stroke 1	Create polygon path with the expected sides, a light stroke weight, and an expected corner condition	Create polygon path with a light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Freedraw, Sides 100, Stroke 50	Create polygon path with the expected sides, medium stroke weight, and an expected corner condition	Create polygon path with a medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance

Polygon, Freedraw, Sides 50, Stroke 50	Create polygon path with the expected sides, medium stroke weight, and an expected corner condition	Create polygon path with a medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Freedraw, Sides 100, Stroke 50	Create polygon path with the expected sides, medium stroke weight, and an expected corner condition	Create polygon path with a medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Freedraw, Sides 2, Stroke 100	Create polygon path with the expected sides, heavy stroke weight, and an expected corner condition	Create polygon path with a heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Freedraw, Sides 50, Stroke 100	Create polygon path with the expected sides, heavy stroke weight, and an expected corner condition	Create polygon path with a heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Freedraw, Sides 100, Stroke 100	Create polygon path with the expected sides, heavy stroke weight, and an expected corner condition	Create polygon path with a heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance

Polygon, Active, Sides 2, Stroke 1	Repeated polygon pattern in selected area with the expected sides, light stroke weight, and an expected corner condition	Repeated polygon pattern in selected area with light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Active, Sides 50, Stroke 1	Repeated polygon pattern in selected area with the expected sides, light stroke weight, and an expected corner condition	Repeated polygon pattern in selected area with light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Active, Sides 100, Stroke 1	Repeated polygon pattern in selected area with the expected sides, light stroke weight, and an expected corner condition	Repeated polygon pattern in selected area with light stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Active, Sides 2, Stroke 50	Repeated polygon pattern in selected area with the expected sides, medium stroke weight expected corner condition	Repeated polygon pattern in selected area with medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Active, Sides 50, Stroke 50	Repeated polygon pattern in selected area with the expected sides, medium stroke weight expected corner condition	Repeated polygon pattern in selected area with medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance

Polygon, Active, Sides 100, Stroke 50	Repeated polygon pattern in selected area with the expected sides, medium stroke weight expected corner condition	Repeated polygon pattern in selected area with medium stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Active, Sides 2, Stroke 100	Repeated polygon pattern in selected area with the expected sides, heavy stroke weight expected corner condition	Repeated polygon pattern in selected area with heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Active, Sides 50, Stroke 100	Repeated polygon pattern in selected area with the expected sides, heavy stroke weight expected corner condition	Repeated polygon pattern in selected area with heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Polygon, Active, Sides 100, Stroke 100	Repeated polygon pattern in selected area with the expected sides, heavy stroke weight expected corner condition	Repeated polygon pattern in selected area with heavy stroke weight	Fail	All “Corner” conditions did not change the Polygon appearance
Star, Anchored, Sides 2, Slope 0.2, Stroke 1	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star’s slope did not always change with the slider’s set numeric value

Star, Anchored, Sides 2, Slope 0.2, Stroke 50	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Anchored, Sides 2, Slope 0.2, Stroke 100	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Anchored, Sides 2, Slope 5, Stroke 1	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Anchored, Sides 2, Slope 5, Stroke 50	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Anchored, Sides 2, Slope 5, Stroke 100	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Anchored, Sides 2, Slope 10, Stroke 1	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value

Star, Anchored, Sides 2, Slope 10, Stroke 5	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Anchored, Sides 2, Slope 10, Stroke 100	Create polygon pattern in selected area with the expected sides, slope, and stroke weight	Create polygon pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Freedraw, Sides 2, Slope 0.2, Stroke 1	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Freedraw, Sides 2, Slope 0.2, Stroke 50	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Freedraw, Sides 2, Slope 0.2, Stroke 100	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value

Star, Freedraw, Sides 2, Slope 5, Stroke 1	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Freedraw, Sides 2, Slope 5, Stroke 50	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Freedraw, Sides 2, Slope 5, Stroke 100	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Freedraw, Sides 2, Slope 10, Stroke 1	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Freedraw, Sides 2, Slope 10, Stroke 5	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value

Star, Freedraw, Sides 2, Slope 10, Stroke 100	Repeated polygon pattern in the trail of the selected area with the expected sides, slope, and stroke weight	Repeated polygon pattern in the trail of the selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Active, Sides 2, Slope 0.2, Stroke 1	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Active, Sides 2, Slope 0.2, Stroke 50	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Active, Sides 2, Slope 0.2, Stroke 100	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Active, Sides 2, Slope 5, Stroke 1	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Active, Sides 2, Slope 5, Stroke 50	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value

Star, Active, Sides 2, Slope 5, Stroke 100	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Active, Sides 2, Slope 10, Stroke 1	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Active, Sides 2, Slope 10, Stroke 5	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight	Fail	The star's slope did not always change with the slider's set numeric value
Star, Active, Sides 2, Slope 10, Stroke 100	Repeated star pattern in selected area with the expected sides, slope, and stroke weight	Repeated star pattern in selected area with the expected sides and stroke weight		The star's slope did not always change with the slider's set numeric value

Tools Tab - Spirograph

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments

Tools Tab - Fill

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
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Fill Color	Fill entire selected area	Fill entire selected area, unless area is selected using the marquee tool, in that case, it may only fill the aforementioned area after it is no longer selectd	Fail	Could not fill areas with a “blank” color.
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Tools Tab - Stamp

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Hypotoid, Inner, Outer, Speed, Resolution				

Tools Tab - Eraser

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Diameter 1, Hardness 1, Flow 1, Opacity 1	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 1, Flow 50, Opacity 1	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 1, Flow 50, Opacity 50	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 1, Flow 50, Opacity 100	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible

Diameter 1, Hardness 1, Flow 100, Opacity 1	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 1, Flow 100, Opacity 50	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 1, Flow 100, Opacity 100	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 50, Flow 1, Opacity 1	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 50, Flow 50, Opacity 1	Remove color values from selected area	Remove color values from selected area	Pass	
Diameter 1, Hardness 50, Flow 50, Opacity 100	Remove color values from selected area	Remove color values from selected area	Pass	
Diameter 1, Hardness 50, Flow 100, Opacity 1	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 50, Flow 100, Opacity 50	Remove color values from selected area	Remove color values from selected area	Pass	
Diameter 1, Hardness 50, Flow 100, Opacity 100	Remove color values from selected area	Remove color values from selected area	Pass	

Diameter 1, Hardness 100, Flow 50, Opacity 1	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 100, Flow 50, Opacity 50	Remove color values from selected area	Remove color values from selected area	Pass	
Diameter 1, Hardness 100, Flow 50, Opacity 100	Remove color values from selected area	Remove color values from selected area	Pass	
Diameter 1, Hardness 100, Flow 100, Opacity 1	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
Diameter 1, Hardness 100, Flow 100, Opacity 50	Remove color values from selected area	Remove color values from selected area	Pass	
Diameter 1, Hardness 100, Flow 100, Opacity 100	Remove color values from selected area	Remove color values from selected area	Pass	
	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
	Remove color values from selected area	Remove color values from selected area	Pass	
	Remove color values from selected area	Remove color values from selected area	Pass	

	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
	Remove color values from selected area	Remove color values from selected area	Pass	
	Remove color values from selected area	Remove color values from selected area	Pass	
	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
	Remove color values from selected area	Remove color values from selected area	Pass	
	Remove color values from selected area	Remove color values from selected area	Pass	
	Remove color values from selected area	Selected content does not appear to be erased	Fail	Selected content may have been erased, but was not visible
	Remove color values from selected area	Remove color values from selected area	Pass	
	Remove color values from selected area	Remove color values from selected area	Pass	

Tools Tab - Picker

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
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Select Fill Color	Selects color in area user chooses and changes the current fill color to the selected color	Selects color in area user chooses and changes the current fill color to the selected color	Pass	Unsure whether “blank”/erased color should be able to be selected
Select Stroke Color	Selects color in area user chooses and changes the current stroke color to the selected color	Selects color in area user chooses and changes the current stroke color to the selected color	Pass	

Supported Devices and Browsers

Test Case Title	Expected Results	Actual Results	Pass/Fail	Comments
Operations on Android Mobile	Full software support	All test case results mirror the those described above	Fail	Additionally to bugs reported, drawing/painting on canvas does not respond
Operations on Windows Laptop	Full software support	All test case results mirror the those described above	Fail	At least one of the test cases did not pass
Operations on Opera GX Browser	Full software support	All test case results mirror the those described above	Fail	At least one of the test cases did not pass
Operations on Google Browsers	Full software support	All test case results mirror the those described above	Fail	At least one of the test cases did not pass
Operations on Microsoft Browsers	Full software support	All test case results mirror the those described above	Fail	At least one of the test cases did not pass

Operations on Apple Browsers (Safari on Mac)	Full software support	All test case results mirror the those described above	Fail	At least one of the test cases did not pass
Operations on Firefox Browser	Full software support	All test case results mirror the those described above	Fail	At least one of the test cases did not pass, passes test cases for saving files

Decision Table Technique

For the first decision table, color combinations were tested. As some of the sliders will readjust themselves, only a few of the settings were selected to minimize the risk of not being able to test a combination. For each selected setting, the upper and lower boundaries were tested. Despite trying to compensate for some combinations being untestable, a few of the entries were unable to be chosen together.

S	100	0	100	0	100	0	100	0	100	0	100	0	100	0	100	0
B	255	255	0	0	255	255	0	0	255	255	0	0	255	255	0	0
R	255	255	255	255	0	0	0	0	255	255	255	255	0	0	0	0
A	100	100	100	100	100	100	100	100	0	0	0	0	0	0	0	0
Result				N/A		N/A		N/A				N/A		N/A		
P/F	P	P	P	N/A	P	N/A	P	N/A	P	P	P	N/A	P	N/A	P	P

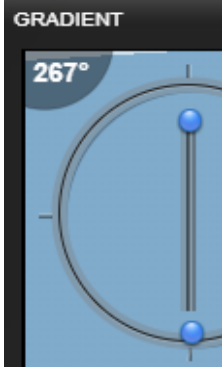
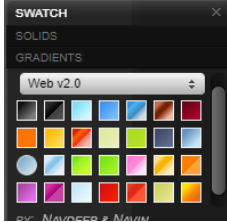
For the second decision table, different colors were tested with diameters and brushes. Blue was created with B set to 255 (R and G at 0) and Red was R at 255 (B and G at 0). Brushes that would produce visible results with the middle diameters 10 and 20 were best suited for these tests.

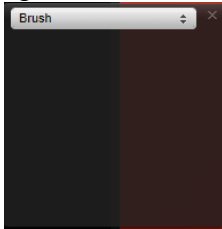
Color	Red	Blue	Red	Blue	Red	Blue	Red	Blue	Red	Blue	Red	Blue
Diamet	10	10	20	20	10	10	20	20	10	10	20	20
Brush	Brush	Brush	Brush	Brush	Hypotrochoid	Hypotrochoid	Hypotrochoid	Hypotrochoid	Calligraphic	Calligraphic	Calligraphic	Calligraphic
Result												

P/F	P	P	P	P	P	P	P	P	P	P	P	P
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Bugs

Test Case	Summary	Error	Steps	Risk Analysis
Color Tab- All Sliders Set At Max	This test case was meant to test the upper boundaries of all sliders under the color tab	The sliders will readjust themselves and make it impossible to have them all at their upper limit	1.Select color tab 2. Slide every slider to be fully right	Low risk, users will likely be able to create their desired color without sliding each slider fully right
Color Tab- All Sliders Set At Middle	This test case was meant to test a middle ground between boundaries for the color sliders	The sliders will readjust themselves and make it impossible to have them all in the middle	1.Select color tab 2. Slide every slider to their middle position	Low risk, users will likely be able to create their desired color without sliding each slider to the middle
Gradient Tab- Circle Slider Top of Circle	This test case was meant to test a boundary for the gradient direction slider	The top of the circle read at a 92 degrees instead of the actual 90 degrees it was at 	1.Select gradient tab 2.Slide circle slider to be at top position	Low risk, the tool is still functioning as expected, only the presentation of degrees are incorrect

Gradient Tab- Circle Slider Bottom of Circle	This test case was meant to test a boundary for the gradient direction slider	The top of the circle read at a 267 degrees instead of the actual 270 degrees it was at 	1.Select gradient tab 2.Slide circle slider to be at bottom position	Low risk, the tool is still functioning as expected, only the presentation of degrees are incorrect
Swatch Tab- Gradients	This test case was meant to test the presentation of the gradient swatches	Some of the gradient swatches weren't gradients at all, instead they were solid colors 	1.Select swatch tab 2.Select the Gradients option 3. View swatches	Low risk, the gradient swatches are still presented and the solid options are just unnecessary extras
History Tab- Titles	This test case was meant to test the functionality of the titled history of operations	The operations were titled incorrectly and seemingly random	1.Select the history tab 2.Make a series of 15+ operations and verify the the titles for them	High risk, the history tab is not functioning as it is meant to, can cause users confusion and hinder their ability to use the software

Options Tab- Calligraphy Selector Diameter Slider Min (1)	This test case was meant to test the lower boundary of the calligraphy diameter	The calligraphy is not visible at its diameter of 1	1.Select options tab 2.Click dropdown and select Calligraphy 3.Slide diameter down to 1	Low risk, small calligraphy diameters are still obtainable for users
Options Tab- Brush All Operations	These test cases were meant to test the functionality of the brush tool	Originally everything worked as intended but, unknowingly how, the tab stopped presenting its options 	1.Select options tab 2.Click dropdown and select Brush	High risk, the functionality of the brush tool is complete unavailable and the operations made to present such a bug is unknown
Save Button- All Operations	These test cases were meant to test the functionality of the save button and the files which it would save	Save button is unresponsive and most cases were unable to test	1.Click save button	High risk, the functionality of the save button is bugged and users are unable to save their work

Participation

Mia: Test cases, equivalence partitions, boundary conditions, for majority of tools, testing approach

Dorothy: Test cases, equivalence partitions, boundary conditions, and bug reports for horizontal tabs on the top of the software, half of browser/device test cases, decision tables one and two